

Estimate calculations by rounding

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000. Copy or complete the first step.

2. Estimate $8,102 \times 49$ using convenient rounded numbers.

3. Estimate $623,590 - 188,420$ to the nearest 100,000.

4. Miro says: Miro believes an estimate must equal the exact answer. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000. Copy or complete the first step.

Answer $400,000 + 200,000 = 600,000$.

2. Estimate $8,102 \times 49$ using convenient rounded numbers.

Answer $8,000 \times 50 = 400,000$.

3. Estimate $623,590 - 188,420$ to the nearest 100,000.

Answer $600,000 - 200,000 = 400,000$.

4. Miro says: Miro believes an estimate must equal the exact answer. Is this correct? Explain.

Answer An estimate is deliberately approximate and should show the likely size of the exact answer.

Estimate calculations by rounding

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000.

6. Check this result using an inverse, estimate or known fact: Estimate $623,590 - 188,420$ to the nearest 100,000.

2. Estimate $8,102 \times 49$ using convenient rounded numbers.

7. Choose an efficient estimate for $48,912 \times 198$ and explain the accuracy.

3. Estimate $623,590 - 188,420$ to the nearest 100,000.

8. Identify and correct this misconception: Miro believes an estimate must equal the exact answer.

4. Solve this independently and show every stage: Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Estimate $8,102 \times 49$ using convenient rounded numbers.

10. Write and solve a similar question to: Estimate $623,590 - 188,420$ to the nearest 100,000.

Teacher answers and checking prompts

1. Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000.
Answer $400,000 + 200,000 = 600,000$.
2. Estimate $8,102 \times 49$ using convenient rounded numbers.
Answer $8,000 \times 50 = 400,000$.
3. Estimate $623,590 - 188,420$ to the nearest 100,000.
Answer $600,000 - 200,000 = 400,000$.
4. Solve this independently and show every stage: Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000.
Answer $400,000 + 200,000 = 600,000$.
5. Use a second method or representation for: Estimate $8,102 \times 49$ using convenient rounded numbers.
Answer Expected result: $8,000 \times 50 = 400,000$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Estimate $623,590 - 188,420$ to the nearest 100,000.
Answer Expected result: $600,000 - 200,000 = 400,000$. A valid check must be shown.
7. Choose an efficient estimate for $48,912 \times 198$ and explain the accuracy.
Answer About $50,000 \times 200 = 10,000,000$. Other justified close estimates are acceptable.
8. Identify and correct this misconception: Miro believes an estimate must equal the exact answer.
Answer An estimate is deliberately approximate and should show the likely size of the exact answer.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Estimate $623,590 - 188,420$ to the nearest 100,000.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Estimate calculations by rounding

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Choose an efficient estimate for $48,912 \times 198$ and explain the accuracy.

6. Create a new example that tests the same idea as: Estimate $623,590 - 188,420$ to the nearest 100,000.

2. Prove the answer to this question, rather than only calculating it: Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Estimate $8,102 \times 49$ using convenient rounded numbers.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $600,000 - 200,000 = 400,000$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro believes an estimate must equal the exact answer.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Choose an efficient estimate for $48,912 \times 198$ and explain the accuracy.
Answer About $50,000 \times 200 = 10,000,000$. Other justified close estimates are acceptable.
2. Prove the answer to this question, rather than only calculating it: Estimate $398,742 + 203,118$ by rounding each number to the nearest 100,000.
Answer Expected result: $400,000 + 200,000 = 600,000$. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Estimate $8,102 \times 49$ using convenient rounded numbers.
Answer Expected result: $8,000 \times 50 = 400,000$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $600,000 - 200,000 = 400,000$.
Answer One valid reconstruction is: Estimate $623,590 - 188,420$ to the nearest 100,000. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro believes an estimate must equal the exact answer.
Answer An estimate is deliberately approximate and should show the likely size of the exact answer.
6. Create a new example that tests the same idea as: Estimate $623,590 - 188,420$ to the nearest 100,000.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.